



AI ASSISTANT USAGE IN STUDENT LIFE

Full-stack Business Intelligence solution

AI ASSISTANT USAGE IN STUDENT LIFE

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Table of Contents

Contents

Table of Contents	2
1. Data Overview	3
1.1 Analytical Questions	9
2. Conceptual, Logical & Physical Model	11
2.1 Logical Model	12
2.2 Physical Model	13
3. SSIS Overview	14
4. SSAS Overview	Error! Bookmark not defined.
5. Power BI Overview	17

1. Data Overview

The project is based on a AI Assistance Usage in Student Life Dataset comprising 10,000 rows. The dataset is used to build a full-stack Business Intelligence solution using Python for EDA, SQL for Schema Creation, SSIS for ETL, SSAS Tabular for analytics, and Power BI for interactive reporting.

Data Source

- Input Format: CSV file
- Columns: SessionID, StudentLevel, Discipline, SessionDate, SessionLengthMin, TotalPrompts, TaskType, AI_AssistanceLevel, FinalOutcome, UsedAgain, SatisfactionRating

Data Dictionary / Catalog

Column Name	Description
SessionID	Unique identifier for each student session
StudentLevel	Student's academic level
Discipline	Academic subject area
SessionDate	Date when the session occurred
SessionLengthMin	Total session duration in minutes
TotalPrompts	Number of AI prompts/questions generated during the session
TaskType	Type of learning task
AI_AssistanceLevel	Level of AI assistance provided
FinalOutcome	Final result of the session/assignment
UsedAgain	Whether the student returned for another session
SatisfactionRating	Student satisfaction rating after session

Data Profiling

1) SessionID

Attribute	Details
Classification	IDENTIFIER
Data Type	str
Unique Values	10000
Missing Values	0 (0.0%)
Sample Values	SESSION00001, SESSION00002, SESSION00003
String Checks	Empty: 0, Whitespace: 0
Casing	100% Uppercase
Length	Min: 12, Max: 12, Avg: 12.00
Action	DROP for modeling (use as index only)

2) StudentLevel

Attribute	Details
Classification	CATEGORICAL
Data Type	str
Unique Values	3
Missing Values	0 (0.0%)
Sample Values	Undergraduate, Undergraduate, Undergraduate
String Checks	Empty: 0, Whitespace: 0
Casing	100% Title Case
Length	Min: 8, Max: 13, Avg: 11.60
Action	One-hot or frequency encode if the endgoal is AI training

3) Discipline

Attribute	Details
Classification	CATEGORICAL
Data Type	str
Unique Values	7
Missing Values	0 (0.0%)
Sample Values	Computer Science, Psychology, Business
String Checks	Empty: 0, Whitespace: 0
Casing	100% Title Case
Length	Min: 4, Max: 16, Avg: 9.01
Action	One-hot or frequency encode if the endgoal is AI training

4) SessionDate

Attribute	Details
Classification	TEXT_DATE
Data Type	str
Unique Values	366
Missing Values	0 (0.0%)
Sample Values	2024-11-03, 2024-08-25, 2025-01-12
Date Range	2024-06-24 to 2025-06-24
Future Dates	0
Format Issues	0 (100% valid)
Timeline Gaps	0 (continuous daily coverage)
Action	Convert to datetime, extract features

5) SessionLengthMin

Attribute	Details
Classification	CONTINUOUS
Data Type	float64
Unique Values	4078
Missing Values	0 (0.0%)
Sample Values	31.2, 13.09, 19.22
Min	0.03
Max	110.81
Mean	19.85
Median	16.65
Std Dev	13.9
Zero Count	0
Negative Count	0
Skewness	1.35 (Right-skewed)
IQR Outliers	302 (Upper bound: 52.23)
Action	Scale & check outliers (cap at 52.23)

6) TotalPrompts

Attribute	Details
Classification	COUNT_DATA
Data Type	int64
Unique Values	37
Missing Values	0 (0.0%)
Sample Values	11, 6, 5
Min	1
Max	39
Mean	5.61
Median	4

Std Dev	4.65
Zero Count	0
Negative Count	0
Skewness	1.77 (Highly Right-skewed)
IQR Outliers	265 (Upper bound: 17.00)
Action	Poisson/Negative Binomial models

7) TaskType

Attribute	Details
Classification	CATEGORICAL
Data Type	str
Unique Values	6
Missing Values	0 (0.0%)
Sample Values	Studying, Studying, Coding
String Checks	Empty: 0, Whitespace: 0
Casing	100% Title Case
Length	Min: 6, Max: 13, Avg: 8.52
Action	One-hot or frequency encode

8) AI_AssistanceLevel

Attribute	Details
Classification	ORDINAL
Data Type	int64
Unique Values	5
Missing Values	0 (0.0%)
Sample Values	2, 3, 3
Min	1
Max	5
Mean	3.48
Median	4
Std Dev	0.99
Zero Count	0
Negative Count	0
Skewness	-0.24 (Slightly Left-skewed)
IQR Outliers	241 (Lower bound: 1.50)
Action	Preserve order (ordinal encoding)

9) FinalOutcome

Attribute	Details
Classification	CATEGORICAL
Data Type	str
Unique Values	4
Missing Values	0 (0.0%)
Sample Values	Assignment Completed, Assignment Completed, Assignment Completed
String Checks	Empty: 0, Whitespace: 0
Casing	100% Title Case
Length	Min: 7, Max: 20, Avg: 14.79
Action	One-hot or frequency encode

10) UsedAgain

Attribute	Details
Classification	BINARY
Data Type	bool
Unique Values	2
Missing Values	0 (0.0%)
Sample Values	True, True, True
Action	Use as 0/1 directly

11) SatisfactionRating

Attribute	Details
Classification	CONTINUOUS
Data Type	float64
Unique Values	41
Missing Values	0 (0.0%)
Sample Values	1.0, 2.0, 3.3
Min	1
Max	5
Mean	3.42
Median	3.5
Std Dev	1.14
Zero Count	0
Negative Count	0
Skewness	-0.34 (Slightly Left-skewed)
IQR Outliers	0 (No outliers)
Action	Scale & check outliers

Data Issues Found During EDA

Issue Description	Suggested Way On How i will Solve it
In SessionID the word 'SESSION' should be removed from it	Remove the SESSION word from each row in this dataset
SessionID Data Type is str	Change the data type to int or index optimized data type
SessionDate Data Type is str	Change the data type to date
AI_AssistanceLevel Column Header is not standardized in terms of the formatting like other columns and contains a special charater that is used for readability	Choose a formatting and use it on all of the column headers.
Consistency Check shows us outliers unrealistic speed	Nothing, Sadly we can't confirm if it is unrealistic speed, we would have needed extra columns for that shows us the length of each prompt. This case could happen if a new user is trying the AI tool for the first time and prompting 'Hi'

1.1 Analytical Questions

Q1: Session Length Optimization

"What is the optimal session length (SessionLengthMin) that maximizes satisfaction rating (SatisfactionRating) for students of different levels (StudentLevel), to identify the ideal session duration for each student segment?"

Q2: Task Type & AI Effectiveness

"How does the effectiveness of AI_AssistanceLevel vary across different TaskTypes (6 categories) in predicting FinalOutcome, to prioritize AI feature development for the most impactful task types?"

Q3: Subject Difficulty & Performance

"Which Disciplines (7 categories) show the strongest relationship between TotalPrompts and FinalOutcome, to identify subjects where students need the most support and where prompt-based learning is most effective?"

Q4: Prompt Usage & Outcomes

"What is the relationship between TotalPrompts and FinalOutcome, controlling for AI_AssistanceLevel, to identify the optimal number of prompts that maximizes completion rates without causing cognitive overload?"

Q5: Time-Based Performance Patterns

"How does student performance (FinalOutcome) and engagement (SessionLengthMin, TotalPrompts) vary by day of week and month (SessionDate), to identify optimal timing for interventions and resource allocation?"

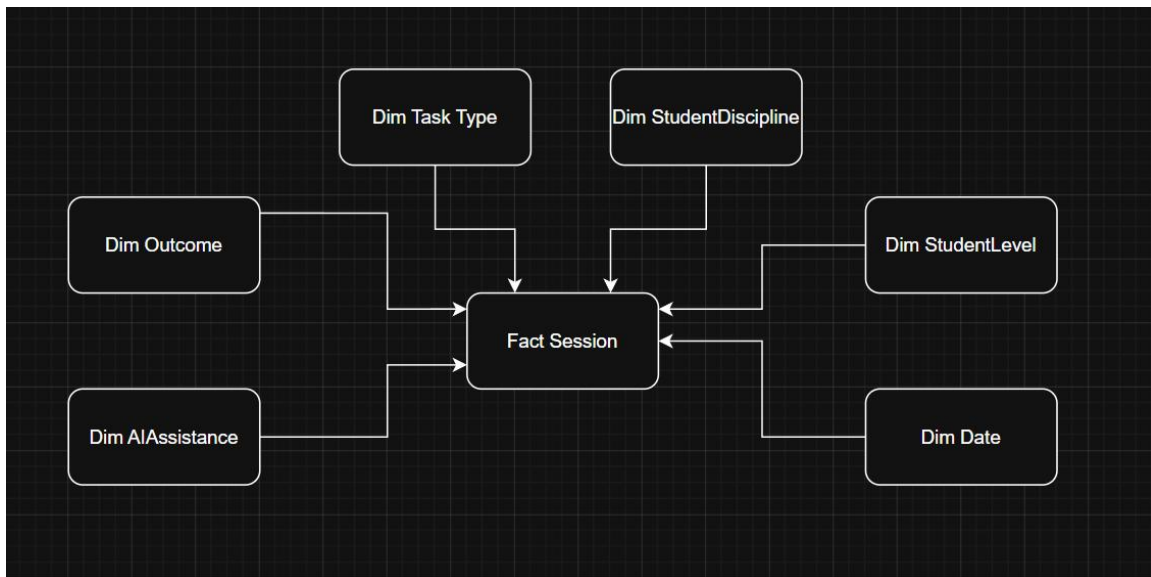
Q6: AI Assistance & Satisfaction Drivers

"What is the combined effect of AI_AssistanceLevel, SessionLengthMin, and TotalPrompts on SatisfactionRating, and which of these factors has the strongest influence, to prioritize improvements for maximum satisfaction impact and to create a student risk score for early intervention?"

2. Conceptual, Logical & Physical Model

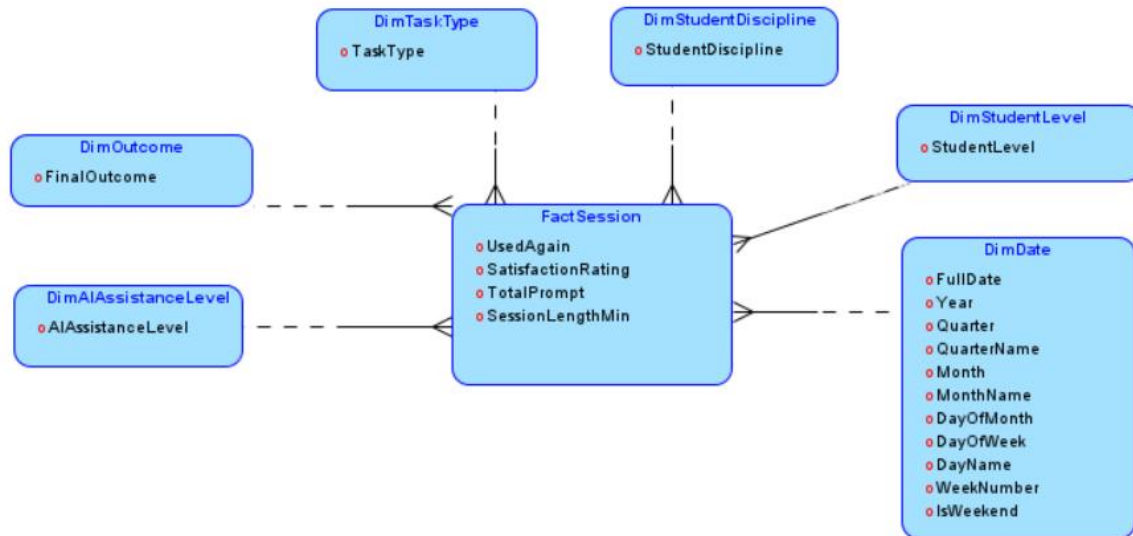
2.1 Conceptual Model

The conceptual model defines the core entities, their roles, and relationships within the business context—abstracted away from technical implementation. It describes what data exists (Tables) and how it's related, without getting into physical storage or data types.



2.2 Logical Model

A logical model is a detailed, technology-independent representation of the data or information requirements of a system or business domain. It describes what data is needed and how it is structured, without specifying how it will be physically implemented.



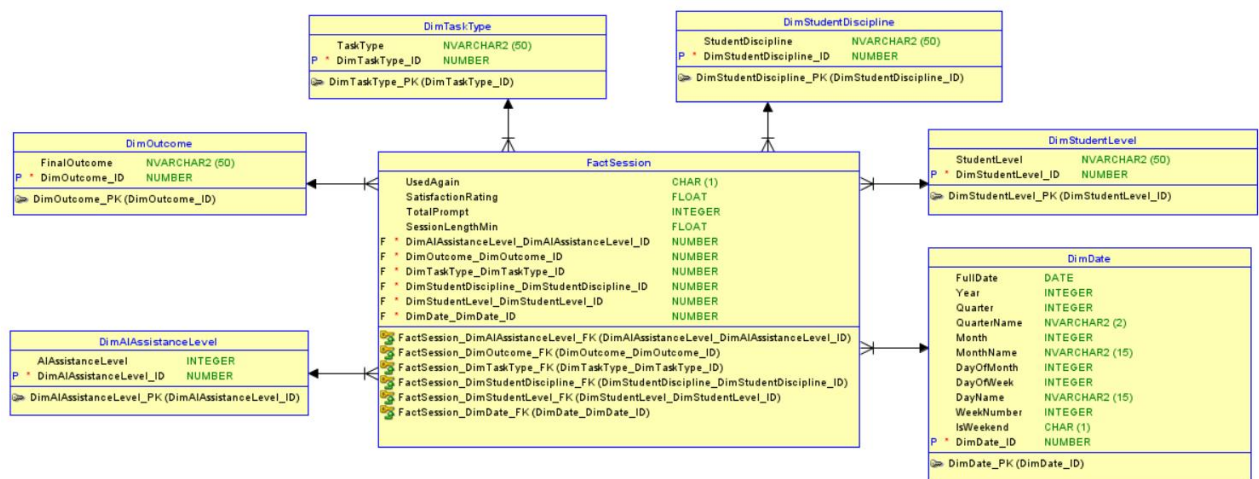
2.3 Relation Model

Tables Stored Physically:

- FactSession (Fact table storing measurable metrics: satisfaction, session length, prompts, reuse)
- DimDate (Calendar date attributes with pre-populated 2020–2030 range)
- DimStudentLevel (Student academic level classifications)
- DimDiscipline (Academic discipline categories)
- DimOutcome (Final session outcome statuses)
- DimTaskType (Task type classifications)
- DimAIAssistanceLevel (AI assistance intensity levels)

Structure:

- Surrogate keys (NUMERIC(28) with IDENTITY) used for all dimension and fact tables
- Foreign key relationships defined between FactSession and all six dimensions
- Star schema with one central fact surrounded by lookup dimensions, optimized for analytical querying
- Error schema with dedicated lookup failure tables for ETL error handling
- Clustered primary keys on all tables; DimDate pre-seeded via system object cross join



3. SSIS Overview

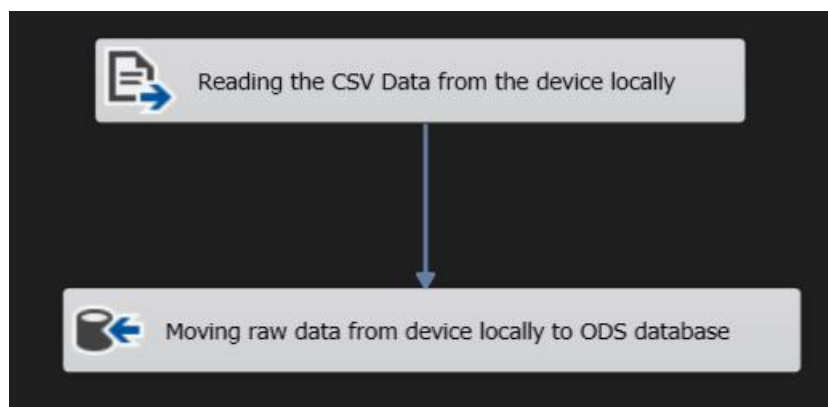
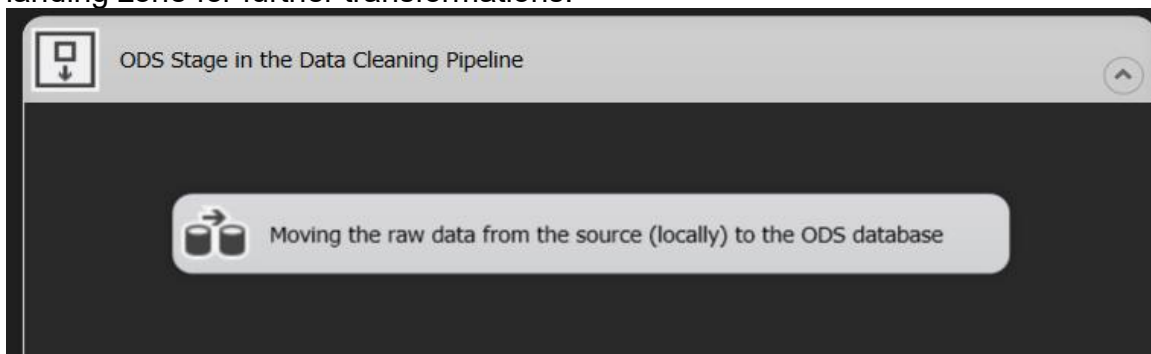
To build a clean, high-quality data model for analysis, a robust ETL (Extract, Transform, Load) pipeline was implemented using SQL Server Integration Services (SSIS). The pipeline processes raw flat files, performs necessary transformations and data cleansing, and loads the refined data into a structured Data Warehouse (DWH).

The ETL process is divided into three main layers:

- ODS (Operational Data Store)
- Staging Area
- Data Warehouse (DWH)

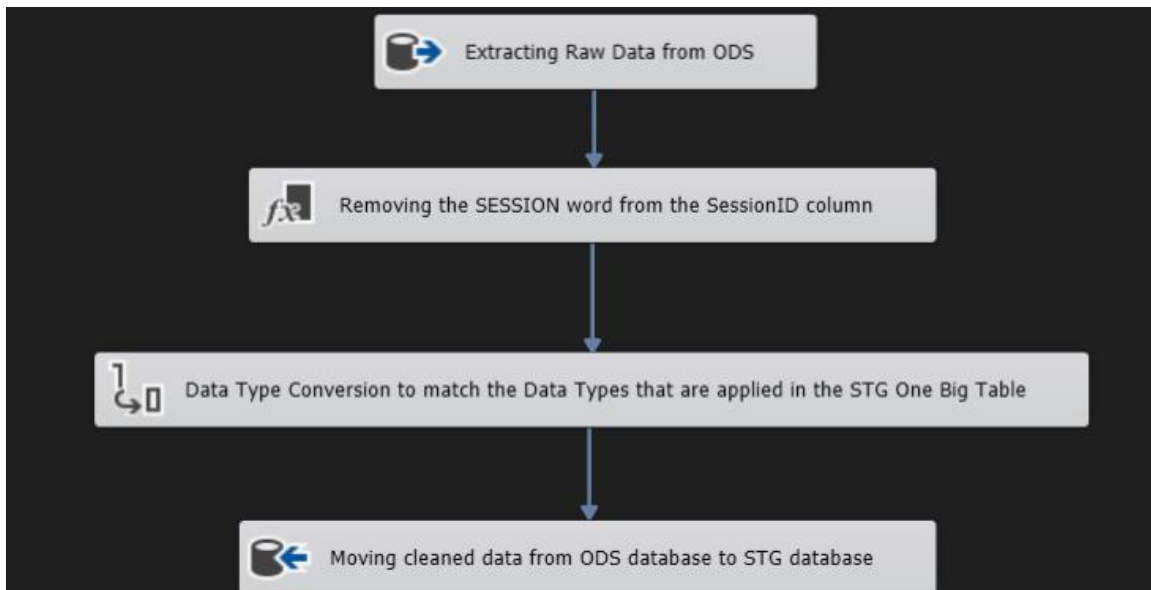
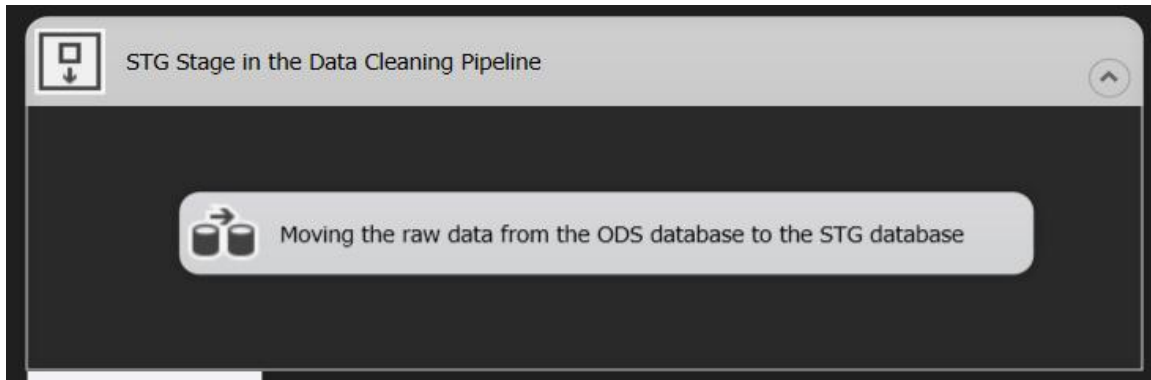
1. Operational Data Store (ODS)

In the ODS layer, data from the 1 source CSV files was loaded using Flat File Source components. This raw table preserve the original format and act as a landing zone for further transformations.



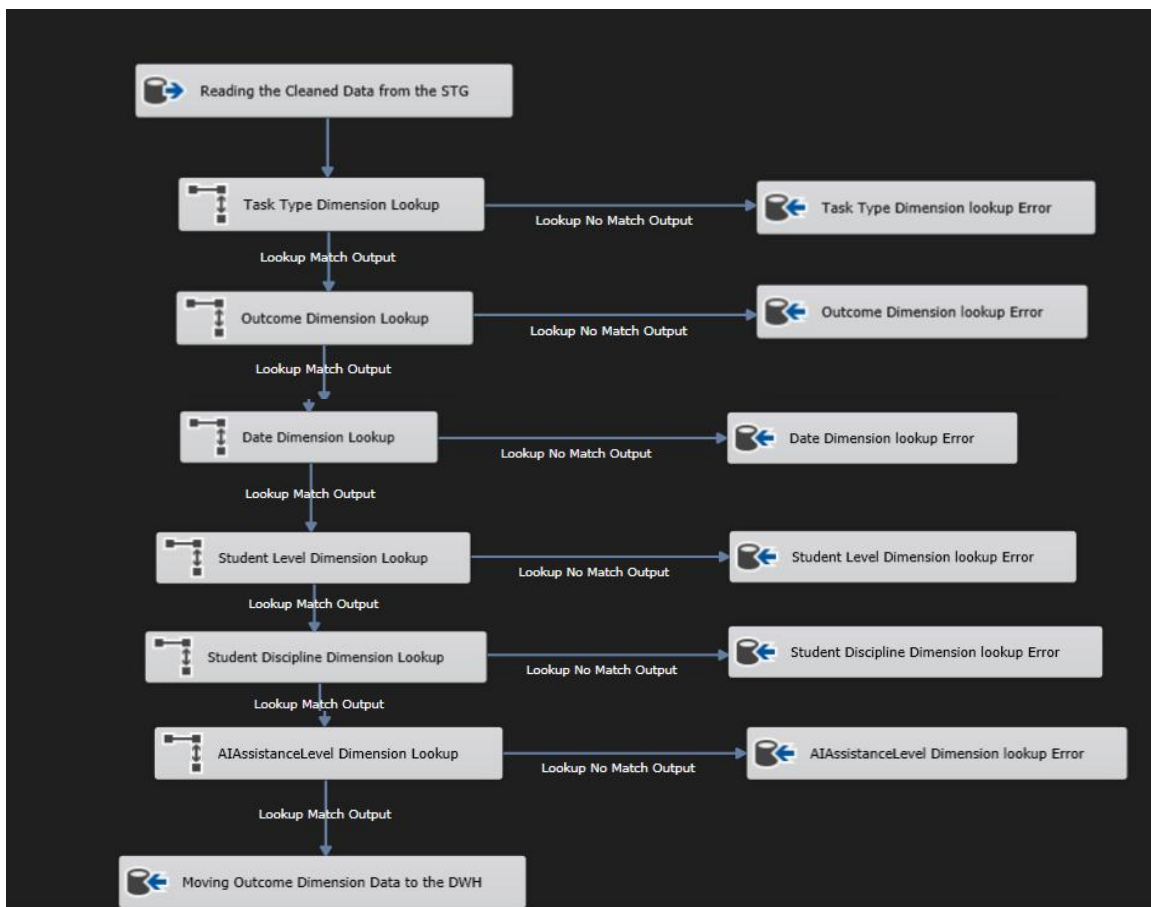
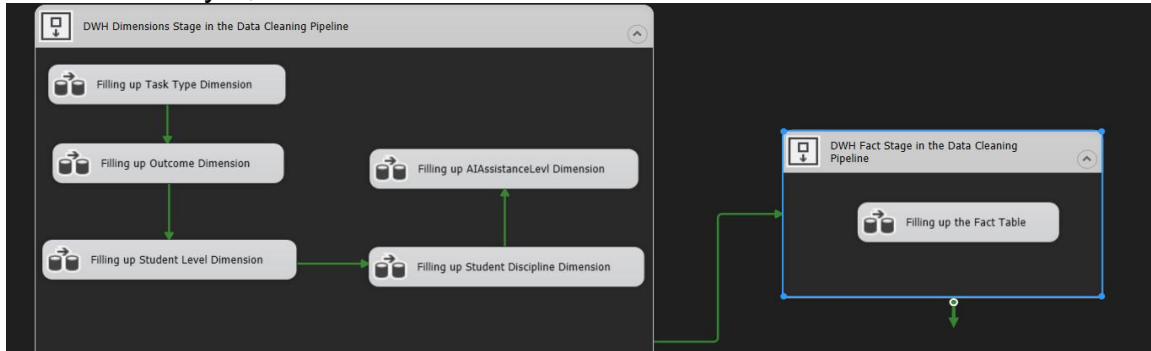
2. Staging Area (STG)

In the Staging layer, data was transformed and cleaned to prepare it for loading into the final warehouse schema. We are still using the One Big Table Design Pattern:



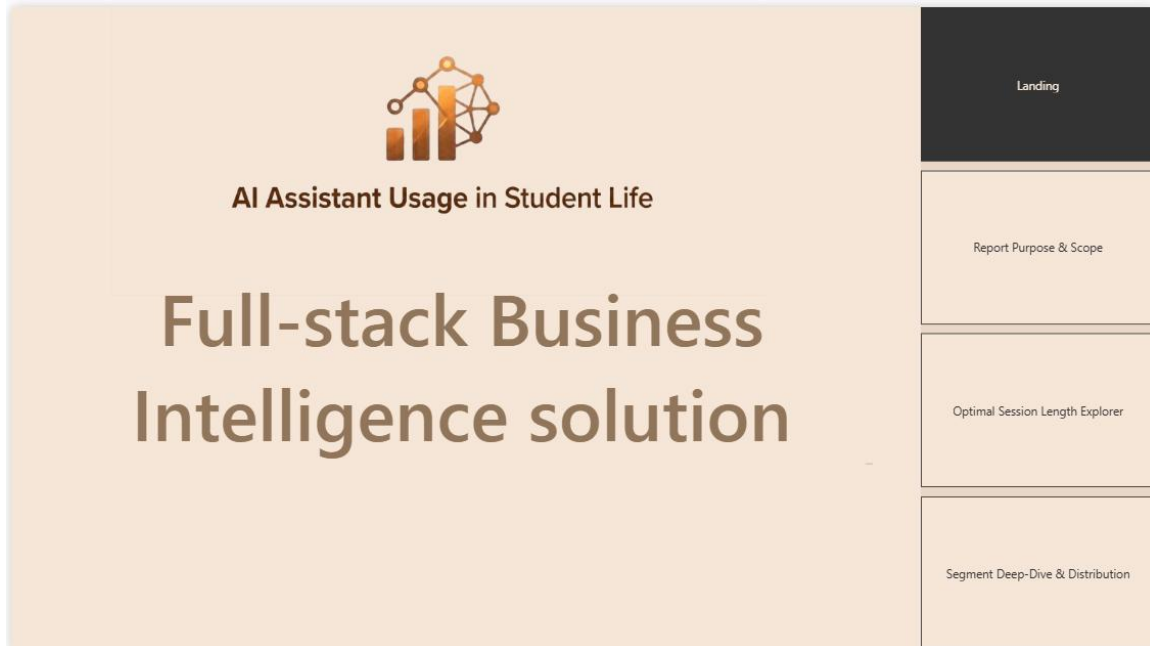
3. Data Warehouse (DWH)

The final step in the ETL process was loading the clean data into the Data Warehouse layer, which follows a star schema structure.

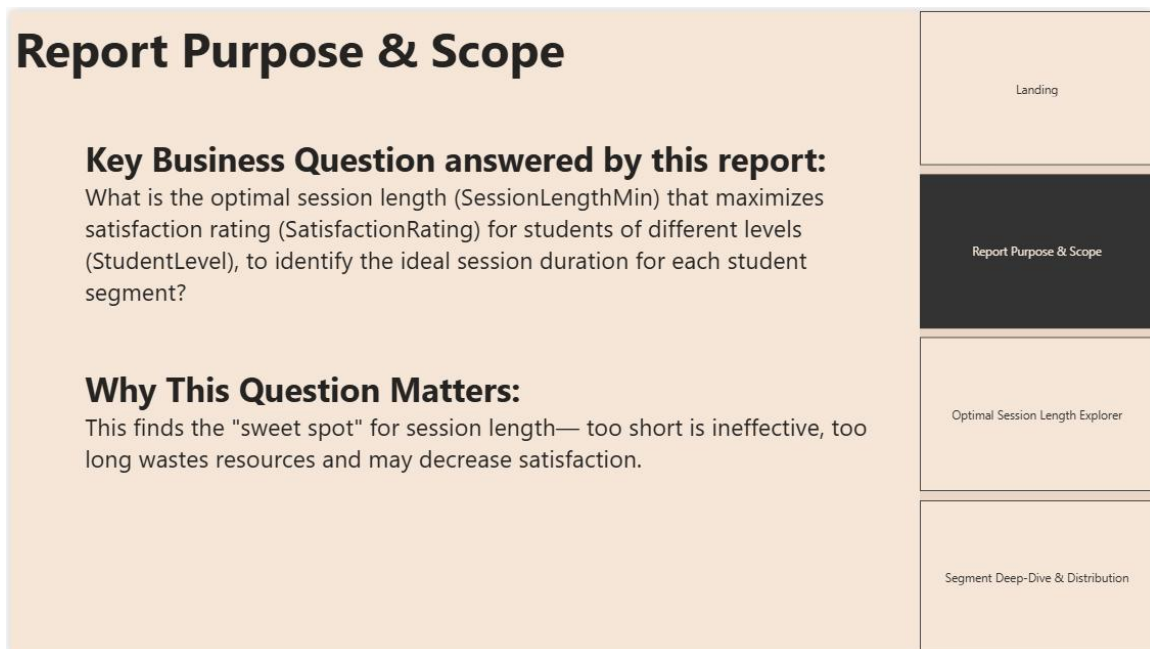


4. Power BI Overview

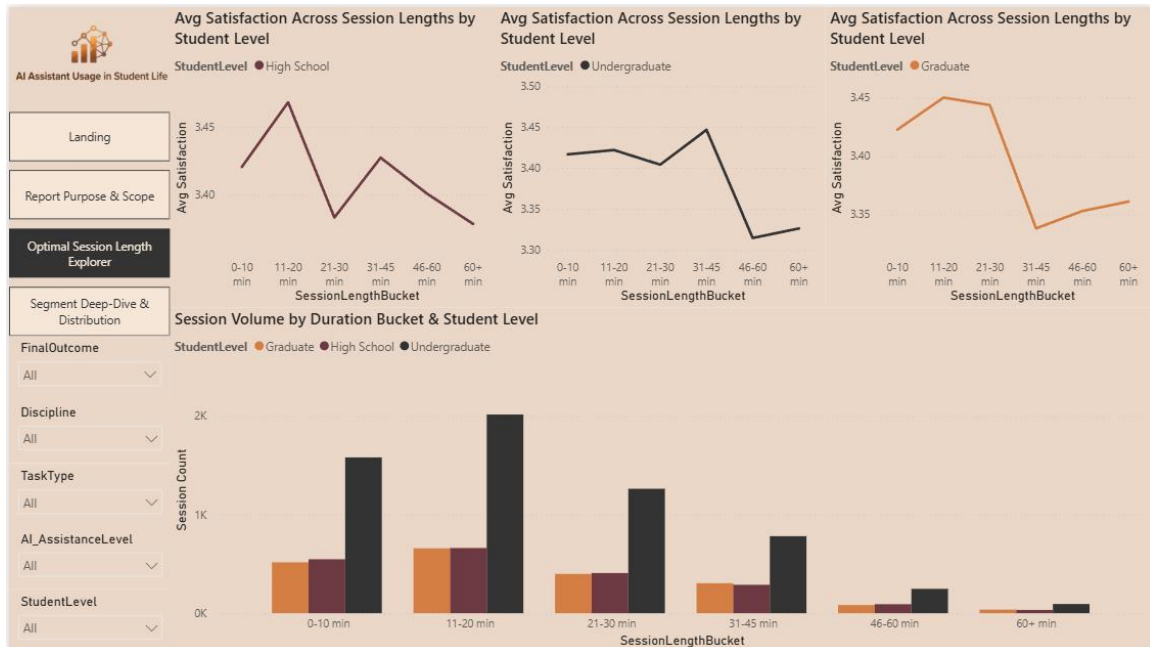
Page 1: Landing Page



Page 2: Report Purpose & Scope



Page 3: Optimal Session Length Explorer



Page 4: Segment Deep-Dive & Distribution

